

Toward Robust Assistive Robot Navigation with EEG Motor Imagery

Mohamed A. Hanafy, Ashutosh Prakash, Saykhun Yusufjonov, Madan M. Rayguru, and Dan O. Popa

Next Generation Systems Group (NGS) | Louisville Automation & Robotics Research Institute (LARRI) | University of Louisville, Kentucky, USA

Overview of the Proposed Framework

A robustness-oriented EEG decoding architecture for cooperative robot navigation

We address inter-session drift and inter-person variability in EEG motor imagery, and integrate the decoder with confidence-aware decision logic and shared autonomy for safe, effective navigation.

1. EEG Variability Across Sessions and Users

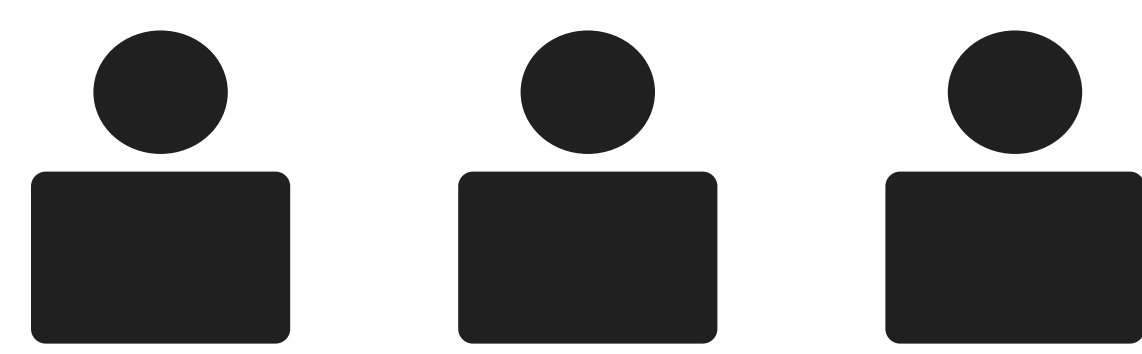


EEG acquisition

15-channel sensorimotor
EEG
125 Hz

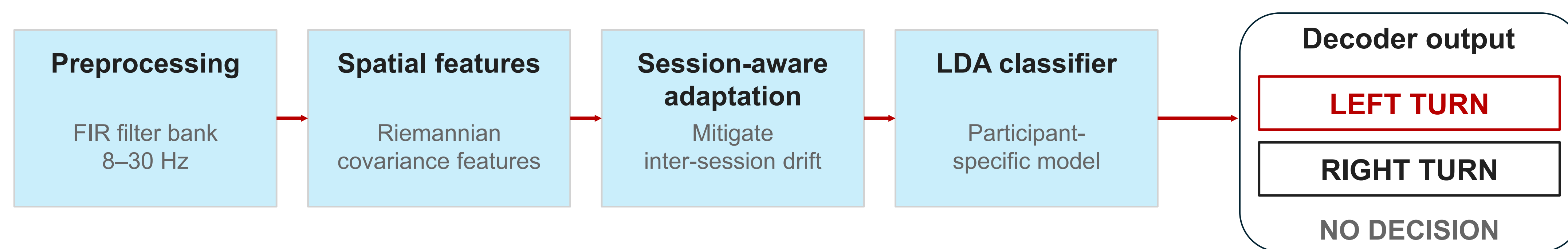


Signals vary across sessions with fatigue, attention, time of day, and physiology.



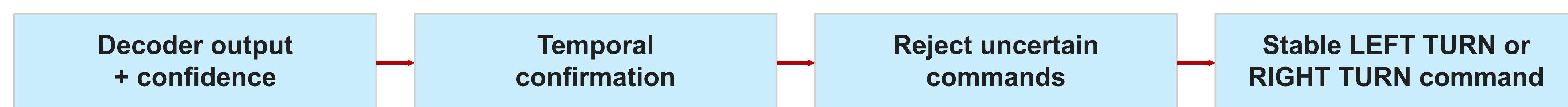
Inter-person variability
Different users, different patterns

2. Robust EEG Intent Decoder



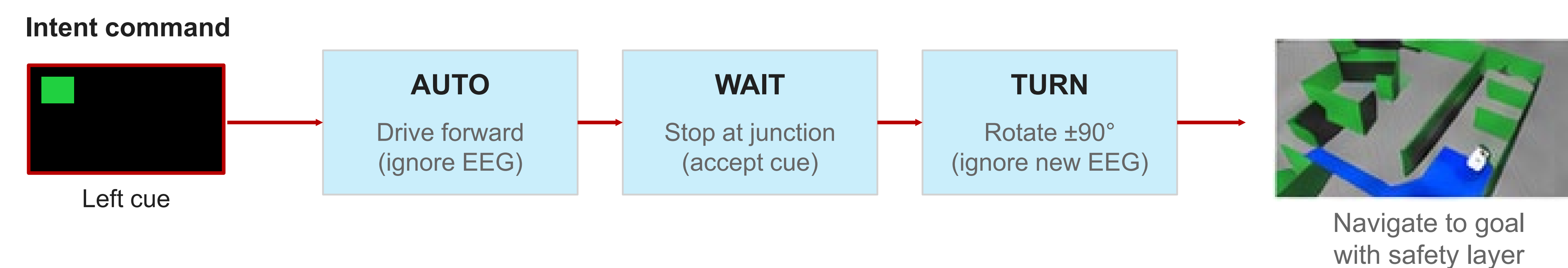
Current scope: binary LEFT TURN / RIGHT TURN classification. NO DECISION is an abstention.

3. Confidence-Aware Decision Logic



Confidence rejection and temporal confirmation reduce uncertain commands before motion execution.

4. Shared Autonomy for Robot Navigation



Shared autonomy checks motion safety and can reject or override unsafe actions.

Why Robustness Matters

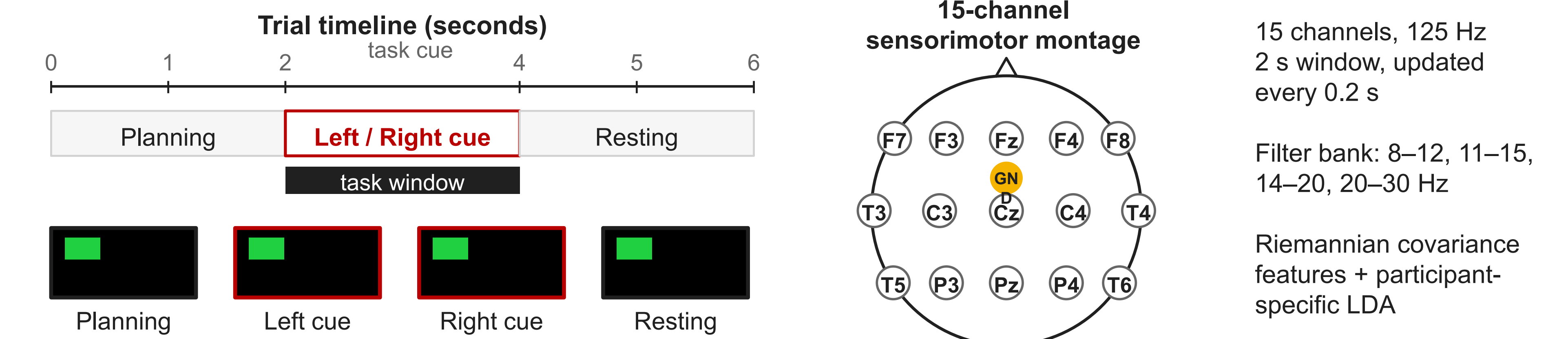
EEG motor imagery can provide a hands-free communication channel. Signal drift across sessions and variability across users limit practical use. Fatigue, time of day, attention, and individual neurophysiology can shift the EEG distribution and reduce decoder reliability.

Robust decoding supports reliable, safe human–robot interaction in real-world use.

Main Contribution

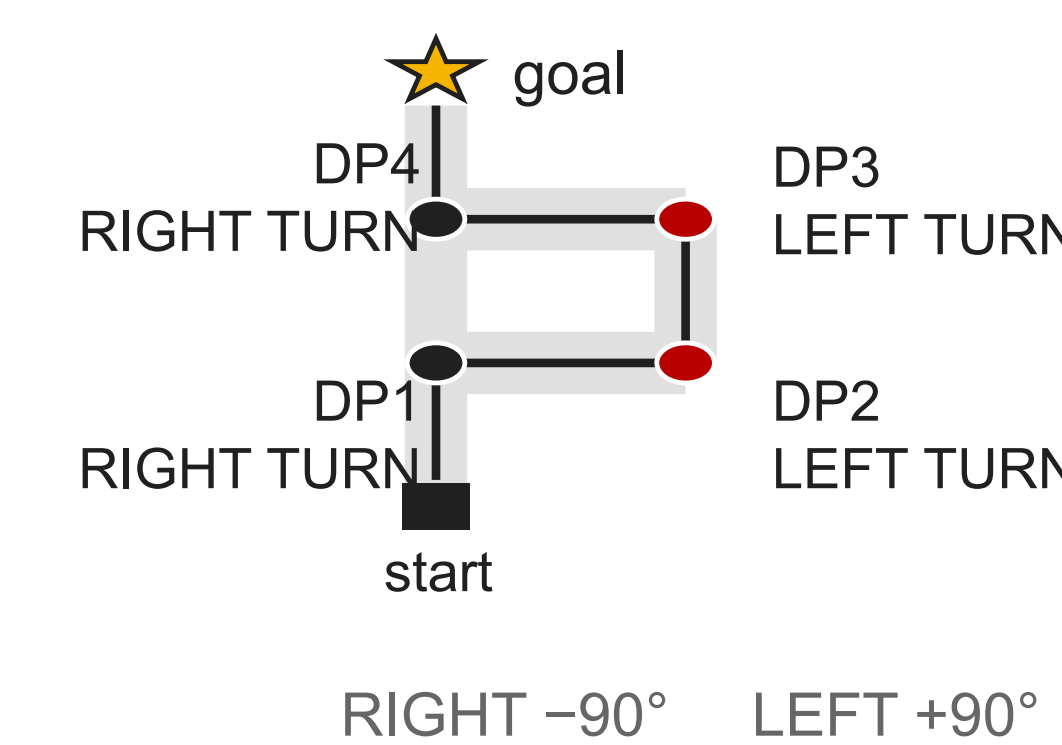
- A robustness-oriented EEG decoding architecture for motor imagery
- Designed to handle inter-session drift and inter-person variability
- Covariance features, session-aware adaptation, confidence rejection, and temporal confirmation
- Cooperative mobile robot navigation with a shared autonomy safety layer

EEG Decoder

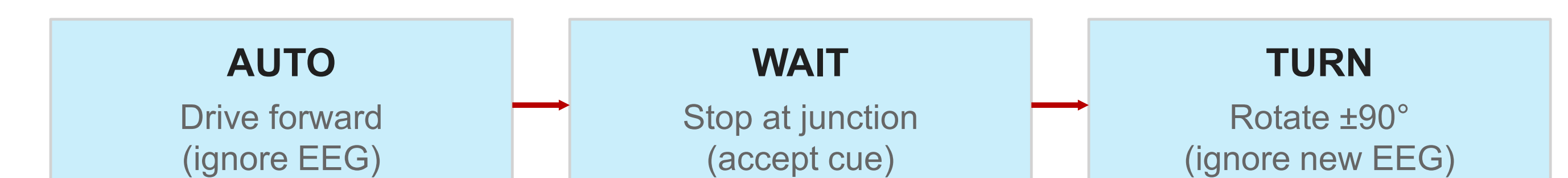


Binary outputs: LEFT TURN or RIGHT TURN; uncertain predictions yield NO DECISION.

Application to Shared Autonomy

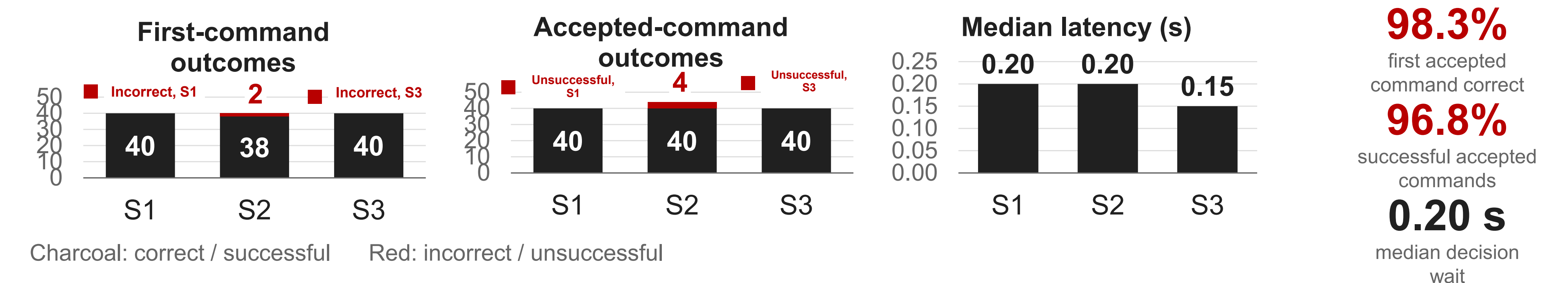


The decoder provides high-level intent. Shared autonomy separates intent estimation from robot motion execution, adds safety checks, and can reject or override unsafe actions.



Experiment and Results

3 participants, 10 traversals each, 4 decision points per traversal, 120 total decision points



Task-level outcomes from the full pipeline, not raw EEG classification accuracy.

Takeaway and Future Research

Reliable intent decoding supports safer human–robot interaction across changing sessions and users. This study uses binary turn decisions; future research can extend the decoder to additional classes. EEG-based control still carries risk, so shared autonomy and safety mechanisms remain essential.